COM3524 Bioinspired Computing

Group Project Marking Criteria

	1st (70-100%)	2:1 (60-69%)	2:2 (50-59%)	3rd Pass or threshold (40-49%)	Fail (30-39%)	Fail (0-29%)
Knowledge & Understanding: The ability to grasp, connect and apply key concepts, theories and terminology within the discipline, demonstrating progression from basic identification to critical synthesis and integration with wider scholarly debates.	Integrates knowledge beyond class material. Synthesises concepts for original insight. Critically engages with debates and wider context.	Explains complex and specialised topics. Uses subject terminology accurately. Connects concepts and explains relevance of debates.	Explains main concepts with some accuracy. Uses key terminology appropriately. Begins to connect ideas and explain debates.	Identifies main concepts from class. Uses basic terminology, sometimes inaccurately. Makes limited or superficial links.	Demonstrates limited understanding of fundamental concepts. Uses basic terminology with frequent inaccuracies. Attempts to make connections, but they are often unclear or incorrect.	Misunderstands or omits key concepts. Uses terminology incorrectly. No connections between ideas.
Critical & Creative Thinking: The ability to construct logical arguments, analyse evidence, and develop solutions, demonstrating progression from basic description to sophisticated evaluation that integrates multiple perspectives and generates original, justified recommendations.	Advances arguments with critical evaluation and original insight. Integrates multiple perspectives for innovative solutions. Critically analyses evidence from several viewpoints. Justifies decisions and makes actionable, creative recommendations.	Logical, substantiated arguments. Synthesises ideas for conclusions. Evaluates evidence from different angles. Develops solutions and recommendations; creative approaches.	Straightforward arguments using taught methods. Connects ideas logically. Reviews and sometimes evaluates evidence. Suggests possible solutions; some creativity.	Basic, limited arguments. Lists ideas with little analysis. Describes evidence with minimal judgement. Identifies standard solutions; little creativity.	Makes simple arguments, but with limited coherence. Presents some ideas, but they may be disjointed or lack development. Limited engagement with evidence; mostly descriptive or inaccurate. Proposes very basic or impractical solutions.	No clear argument or logic. Disconnected or irrelevant ideas. Evidence not explained or unrelated. No problem-solving or creativity.
Communication: The ability to structure and convey ideas clearly through appropriate style, language and format,	Precise, cohesive structure throughout. Style expertly adapted for audience and purpose. Wide range of	Logical, clear structure. Style suits audience and context. Paraphrases and summarises sources well.	Basic, mostly clear structure. Style mostly fits audience. Attempts to paraphrase and summarise. Some technical	Structure is inconsistent. Some awareness of audience and style. Overuses direct quotations. Limited or	Structure is often unclear or fragmented. Limited awareness of audience or purpose, leading to an	No clear structure or logic. Inappropriate style for audience. Source material miscommunicated. Technical language absent or

demonstrating progression from basic coherence to sophisticated adaptation for specific audiences and purposes, with effective integration of sources, technical terminology and accessbility features where relevant.	communication strategies. Seamless integration of sources. Technical language used for precise argumentation.	Accurate use of technical language. Ensures output is accessible to audience.	language, with errors. Uses digital accessbility features with occasional omissions.	inaccurate technical language and use of digital accessbility features.	inappropriate style. Relies heavily on direct quotation, often without proper integration. Minimal or incorrect use of technical language and digital accessbility features.	incorrect. Does not use digital accessbility features.
Practical Skills: The ability to apply discipline-specific technical, creative or artistic competencies, demonstrating progression from basic skill development to innovative and autonomous performance.	Applies specialist skills with innovation and precision. Demonstrates exceptional technical, creative, or artistic mastery.	Performs specialist skills effectively and autonomously. Demonstrates highly developed technical or creative ability.	Applies skills consistently and mostly independently. Completes tasks accurately and proficiently.	Developing basic specialist skills. Completes tasks with some accuracy and independence.	Applies basic specialist skills, but with significant errors or inconsistencies. Requires substantial guidance to complete tasks.	Lacks evidence of skill development or application. Makes frequent errors; little independence.

1. **Knowledge and Understanding:** These should be demonstrated in your review of existing literature and your ability to adapt the pre-existing 2D Game of Life Cellular Automata (CA) to simulate forest fires (presentation and code)
2. **Critical and Creative Thinking:** These should be demonstrated in your ability to adapt and apply the CA to successfully address the client's questions, your ability to critically analyse your simulation results, and your ability to reflect on the limitations of your model (presentation)
3. **Communication:** This will be demonstrated in terms of the quality and clarity of your presentation.
4. **Practical skills:** These will be demonstrated by the code that you submit. Three main features of the code will be considered: functionality, usability (including ease of running code & quality of instructions), and general elegance.
5. **The weighting of these components is as follows**: Knowledge and Understanding: 30%; Critical and Creating Thinking: 30%; Communication: 20%; Practical Skills: 20% (including 10% for functionality of code, 5% for usability of code, and 5% for general elegance of code).

These marks will be adjusted for individual team members using (staff-moderated) BuddyCheck feedback from your peers. Specifically, 100% of your group mark will be multiplied by the capped BuddyCheck factor. On BuddyCheck, you will be asked to rate your peers on different criteria such as overall contribution to the team's work. You will also be asked to add text to justify your marks,

particularly when awarding scores that differ significantly from the average. Note that you **must** provide a clear justification for any particularly high or low score, supported by specific examples of the group member's contribution. Unsupported or insufficiently justified scores may be given less weight or disregarded when individual marks are determined. A buddycheck score will be generated from your responses to these questions and moderated by staff. This score is thresholded (e.g., to ensure factor no less than 0.85, no more than 1.15) then applied to the team score (which again is thresholded to ensure no final score > 100%). The exception is for students that have simply not contributed at all (verified by attendance records for relevant labs) that are awarded a zero.